

THE reaction between benzoyl chloride and aniline (Grant and Hinshelwood, J., 1933, 1351) in hexane solution is rendered anomalous by a marked influence of the solid particles of product, which cause heterogeneous catalysis. In carbon tetrachloride solution the effect is much less, and the reaction is more nearly bimolecular. It now appears that in solvents where the velocity is considerably greater than in carbon tetrachloride or hexane, the homogeneous reaction quite outweighs the heterogeneous, and the anomalies disappear almost entirely. The bimolecular reaction is interesting in that it belongs to the class where the rate of change is much smaller than the possible rate of activation.

$$10^3.N.k = P.Z_{AB}.e^{-E/RT} \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (1)$$

This paper deals with two problems. The first is to determine whether the large influence of substituents in the benzene rings depends chiefly upon changes in E or changes in P , a question which becomes specially important in a reaction where P is so far removed from unity. The second is to explore further the nature of P itself.

The course of the reaction is expressed by the usual bimolecular formula, as shown in Table I, where t is the time in minutes, x the % change, and k the constant (g.-mols./l./sec.); a is the initial concentration of the acid chloride, and b that of the amine in g.-mols./l.; a is always equal to $b/2$. Hence

$$k = \frac{1}{2 \times 60 t} \left\{ \frac{1}{(100 - x)} - \frac{1}{100} \right\} \times \frac{100}{a}$$

The anomalous influence of dilution found with hexane has almost disappeared in benzene. Table II gives the variation of k with dilution. Dilution I represents $a = 0.01$, $b = 0.02$,

TABLE I.

Aniline and benzoyl chloride at 25·0°.			Aniline and <i>p</i> -chlorobenzoyl chloride at 25·0°.			<i>p</i> -Nitroaniline and <i>p</i> -nitrobenzoyl chloride at 70·0°.		
$a = 0·005, b = 0·01.$			$a = 0·0025, b = 0·005.$			$a = 0·005, b = 0·01.$		
$x.$	$t.$	$100k.$	$x.$	$t.$	$100k.$	$x.$	$t.$	$k \times 10^4.$
10	2·5	7·41	20	7·8	10·7	10	210	8·81
20	5·6	7·44	30	13·3	10·7	20	474	8·79
30	9·6	7·45	40	21·1	10·5	30	792	9·01
40	14·6	7·61	50	32·3	10·3	40	1242	8·96
50	21·5	7·75	60	49·7	10·1	50	1920	8·68
60	32·5	7·72	70	76·0	10·2	60	2952	8·47
70	52·2	7·46				70	4680	8·32

dilutions II and III representing one-half and one-quarter of these concentrations respectively. The constants are mean values of series such as those given in Table I.

TABLE II.

$k \times 10^2$ at 25° for various reactants at different dilutions.

	Dilution I.	Dilution II.	Dilution III.
Aniline, benzoyl chloride	7·73	7·55	6·88
<i>p</i> -Chloroaniline, benzoyl chloride.....	1·73	1·63	1·50
Aniline, <i>p</i> -nitrobenzoyl chloride	—	60·7	58·1
<i>p</i> -Toluidine, benzoyl chloride	—	33·4	29·8
<i>m</i> -Nitroaniline, benzoyl chloride	0·0468	0·0444	0·0441

To allow for such small trend as the constant shows, all results, which for practical convenience had to be determined at different dilutions, are corrected to corresponding dilutions by the aid of this table before being compared with one another.

The influence of temperature is given in Table III, the substituents denoting those in the aniline and in the benzoyl chloride; the constants recorded are corrected for the small expansion of benzene with temperature. The values of E are found by plotting $\log k$ against $1/T$ in the usual way.

TABLE III.

Mean values of $k \times 10^2$ for various temperatures.

Substituent.		Dilution.						
Amine.	Acid chloride.		5·0°.	15·0°.	25·0°.	40·0°.	70·0°.	100·0°.
H	<i>p</i> -NO ₂	III	28·6	36·5	58·1	87·1	—	—
<i>p</i> -CH ₃	H	III	13·4	20·7	29·8	55·2	—	—
H	<i>p</i> -Cl	III	4·65	—	10·4	17·8	54·6	—
H	H	II	3·08	—	7·55	12·2	41·7	—
H	<i>p</i> -CH ₃	III	1·64	—	3·91	7·46	23·9	—
<i>p</i> -Cl	H	II	0·68	—	1·63	2·98	9·5	—
<i>p</i> -NO ₂	<i>p</i> -NO ₂	II	—	—	0·0097	0·0214	0·093	0·409
<i>p</i> -NO ₂	H	I	—	—	0·00421	0·0110	0·062	0·253

Influence of Substituents on the Factors determining the Reaction Velocity.—Table IV gives the values of the energy of activation and the values of k , for dilution III, at 25° for all the pairs of compounds investigated.

TABLE IV.

Influence of substituents on E and on the reaction rate.

Substituent.				Substituent.			
Amine.	Acid chloride.	$100k_{25^\circ}$	$E.$	Amine.	Acid chloride.	$100k_{25^\circ}$	$E.$
H	<i>p</i> -NO ₂	58·1	5900	<i>p</i> -Cl	H	1·50	7,600
<i>p</i> -CH ₃	H	29·8	6800	<i>m</i> -NO ₂	H	0·044	10,500
H	<i>p</i> -Cl	10·4	7000	<i>p</i> -NO ₂	<i>p</i> -NO ₂	0·0097	10,400
H	H	6·88	7350	<i>p</i> -NO ₂	H	0·0042	11,800
H	<i>p</i> -CH ₃	3·91	7800				

Qualitatively, the influence of the substituents is what might be predicted from Robinson's theory of the reactivity of organic compounds. The present problem is to decide which of the

factors in equation (1) is chiefly responsible for the changes in velocity. If the substituent influences E principally, leaving P more or less constant throughout the series, then

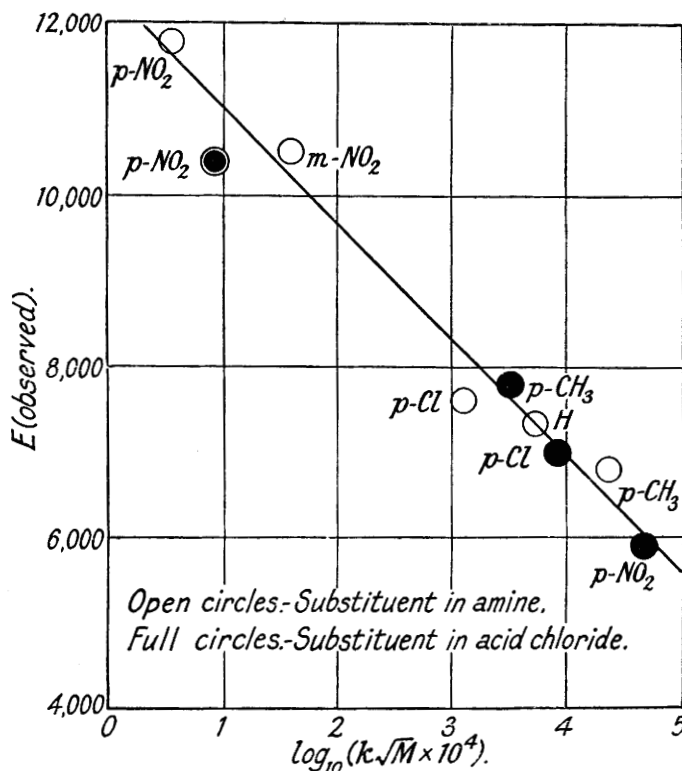
$$\ln(k/Z_{AB}) = \text{constant} - E/RT \quad (2)$$

$Z_{AB} = N^2 \cdot \sigma_{AB}^2 (8\pi RT/M)^{\frac{1}{2}}$, where $1/M = 1/m_A + 1/m_B$, σ_{AB} being the mean diameter, and m_A and m_B the respective molecular weights. For the series of compounds, σ_{AB} will be taken as nearly constant. For comparative purposes at a given temperature, relation (2) reduces to

$$2.303 \log k \cdot M^{\frac{1}{2}} = \text{constant} - E/RT$$

In Fig. 1 the value of $\log(k \cdot M^{\frac{1}{2}} \times 10^4)$ is plotted against the observed value of E . It is important to note that the line is made to pass through the point representing the two unsubstituted compounds, and has the theoretical slope $2.303 RT$ required by the formula.

FIG. 1.



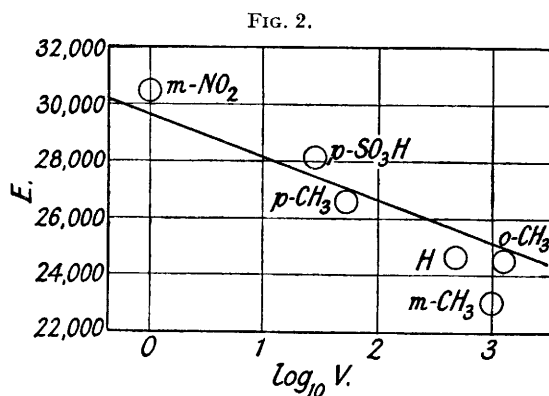
Correlation between variations in velocity and changes in the activation energy.

It is evident that the variations in velocity are in the main accounted for by the changes in the energy of activation. This result is specially interesting in the present example where P is so far removed from unity that very wide variations in it might have been expected. Whatever the nature of P , we may conclude that it remains of the same order of magnitude in any given solvent, and that the very marked internal electronic displacements in the benzene ring, caused by the substituents, may change E considerably, while having far less effect on P . Such variations in P as do occur are not surprising: it is only remarkable that they are not greater.

A similar test can be made for a reaction of quite a different type, *viz.*, one depending upon interaction of solvent and solute, with the results of Cain and Nicoll (J., 1902, 81, 1434) on the decomposition of various diazo-compounds in aqueous solution. Unfortunately, some of the measurements were made over too limited a range of temperature to allow calculation of accurate E values. All that has been done, therefore, is to plot E against the logarithm of the relative speed of decomposition at 60° , as shown in Fig. 2. The line has the theoretical slope of $2.3 \times 1.98 \times 333$. It can be seen that, if changes in E are not the whole effect of the sub-

stituents, they are at least the most important factor. Correction of the E values for variation of the viscosity of water with temperature (see Moelwyn-Hughes, "Kinetics of Reactions in Solution," Oxford, 1933, p. 20) would change the slope slightly, but the data are not accurate enough for elaborate treatment.

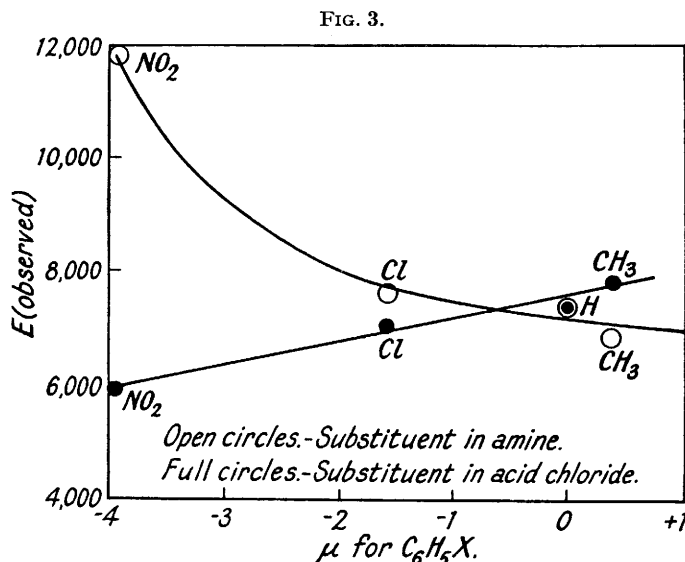
Relation between E and the Polarity of Substituents.—Nathan and Watson (J., 1933, 890,



Values derived from Cain and Nicoll's measurements on decomposition of diazo-compounds.

1248) have shown that for certain reactions of aromatic compounds containing a variable substituent X, $\log(k/Z_{AB})$ shows a functional relation with the dipole moment, μ , of $\text{C}_6\text{H}_5\text{X}$. Changes in $\log(k/Z_{AB})$ were assumed to measure changes in E , i.e., P was assumed constant. Since the reactions dealt with were of the type for which P is usually of the order unity (Grant and Hinshelwood, J., 1933, 258; Moelwyn-Hughes, *op. cit.*), this procedure was reasonable. Nevertheless, it is not without interest to plot directly observed values of E against μ in the same way. This has been done, in Fig. 3, for the two series of reactions, $\text{C}_6\text{H}_5\text{NH}_2$ with $p\text{-X}\cdot\text{C}_6\text{H}_4\cdot\text{COCl}$ and $p\text{-X}\cdot\text{C}_6\text{H}_4\cdot\text{NH}_2$ with $\text{C}_6\text{H}_5\cdot\text{COCl}$, all in benzene solution.

Further Investigation of the Nature of the Factor P .—The rate of reaction is much smaller than that of activating collisions between the reactant molecules. Since, e.g., in nitrobenzene the rate is about 100 times as great as in carbon tetrachloride, it is obvious that



Relation between polarity of substituents and energy of activation.

the solvent plays a fundamental part in determining the interaction. The question arises whether the ratio of the number of molecules reacting to the number of activating collisions is simply measured by the probability that the appropriate solvent molecules should be present during the encounter. Light can be thrown on this question by studying the effect on the reaction in a relatively inert solvent of increasing amounts of a much more active solvent. With benzene as a solvent, the addition of nitrobenzene causes a linear

increase in the velocity of reaction, both for the benzylation of aniline and for that of *m*-nitroaniline. As shown by the results in Table V, the rate can be expressed in the former case by the formula

$$-d[\text{Ph}\cdot\text{COCl}]/dt = k_0[\text{Ph}\cdot\text{COCl}][\text{Ph}\cdot\text{NH}_2] + k_1[\text{Ph}\cdot\text{NO}_2][\text{Ph}\cdot\text{COCl}][\text{Ph}\cdot\text{NH}_2]$$

where k_0 and k_1 at 25° are 0.0755 and 0.0828 respectively.

For *m*-nitroaniline and benzoyl chloride a similar formula holds and k_1 is 11.6×10^{-4} . The form of the equation suggests that the constant k_0 is of the form $k'[\text{C}_6\text{H}_6]$. If we give a kinetic interpretation to the action of the solvent, the linear relationship shows that we are dealing with ternary collisions, and not with collisions of still higher order.

TABLE V.

Additions of nitrobenzene to the benzene solution.

Aniline and benzoyl chloride at 25.0°, dilution II.

Normality of Ph·NO ₂ , <i>n</i>	0	0.2	0.5	0.7	1.0	1.2	1.5	1.7	2.0
100 ($k_0 + k_1n$), obs.	7.55	9.31	11.86	13.51	15.57	17.43	20.14	21.8	24.66
„ calc.	7.55	9.21	11.69	13.34	15.83	17.48	19.96	21.62	24.1

Repetition of the experiments at 5° and at 40° gave a value of 6,500 calories for the *E* corresponding to k_1 .

m-Nitroaniline and benzoyl chloride at 40.0°.

Normality of Ph·NO ₂ , <i>n</i>	0.5	1.0	1.5	2.0
$k_1 \times 10^4 = (k_{\text{total}} - k_0)/n$	10.2	10.5	11.3	11.6

We may now make an estimate of the probability of such collisions. The number of ternary collisions (see *e.g.*, Steiner, *Z. physikal. Chem.*, 1932, *B*, **15**, 249) between molecules A, B, and M per c.c. per sec. is given by the formula

$$Z_{\text{ABM}} = N^3 \tau \sigma_{\text{AB}}^2 \sigma_{\text{ABM}}^2 8\pi RT \sqrt{\frac{1}{m_{\text{A}}} + \frac{1}{m_{\text{B}}}} \sqrt{\frac{1}{m_{\text{AB}}} + \frac{1}{m_{\text{M}}}} C_{\text{A}} C_{\text{B}} C_{\text{M}}$$

where N is Avogadro's number, C_{A} , C_{B} , and C_{M} are the concentrations in g.-mols./c.c., m_{A} , m_{B} , and m_{M} are the molecular weights of A, B, and M, and m_{AB} is that of A and B taken together; σ_{AB} is $(\sigma_{\text{A}} + \sigma_{\text{B}})/2$ and σ_{ABM} is $(\sigma_{\text{A}} + \sigma_{\text{B}} + \sigma_{\text{M}})/2$, where σ_{A} , σ_{B} , and σ_{M} are the collision diameters of A, B, and M. τ is the least definite of the quantities involved, being the duration of a binary collision between A and B. In order not to make arbitrary assumptions about it, a convenient plan will be to calculate the value which it would have to be given for every activated ternary collision to be effective, *i.e.*, in order that P of equation (1) should simply be the probability of the presence of the solvent molecule.

For the reaction represented by the second term of equation (2), we have: no. of g.-mols./l./sec. reacting when all three concentrations are 1 g.-mol./l. = $k_1 = 0.0828$. Therefore the actual number of molecules reacting per c.c. per sec. is $0.0828N \cdot 10^{-3}$. This can be equated to $Z_{\text{ABM}} \cdot e^{-E/RT}$. In calculating Z_{ABM} , $c_{\text{A}} = c_{\text{B}} = c_{\text{M}} = 10^{-3}$. The collision diameters will all be taken as 6×10^{-8} cm.; E is 6500 cal. On this basis, the value of τ is 5×10^{-20} sec. The time during which a pair of colliding molecules are within a molecular distance of one another is 10^{-11} — 10^{-13} sec., unless association occurs, in which case the time will be greater. The value found for τ , therefore, shows that our original assumption is impossible. To give a physically intelligible value to τ , we must assume that a small fraction only, even of the ternary collisions, are effective. Thus the existence of the probability factor, P , independent of activation, cannot, in this example, be explained away as a purely kinetic factor, depending on the necessity for collisions with solvent molecules.

SUMMARY.

The rate of a bimolecular reaction in solution is given by the formula: number of molecules reacting = $P \cdot Z \cdot e^{-E/RT}$, where P is a factor independent of the activation rate. In the benzylation of amines, P is extremely small. In spite of this, it has been found that the influence of substituents on the velocity of benzylation of aromatic amines in

benzene solution depends principally on changes in the activation energy. Changes in P are of much smaller importance as long as the medium is unchanged.

There are indications from other work that changes in E are the important factors determining the variation with substitution of the rate of reaction of aromatic compounds generally.

A functional relation between the energy of activation and the polarity of the substituent groups exists in the benzylation reaction, as well as in other cases, where P approaches unity.

Experiments in mixed solvents indicate that the factor, P , cannot be explained away as a purely kinetic factor depending on ternary collisions with solvent molecules, although such collisions appear to be an essential part of the reaction mechanism.

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